Video Streaming Configuration

Video Streaming Configuration

This guide covers the configuration of video streaming for devices in GCXONE, including cloud and local mode setup, stream quality settings, and optimization options.

Estimated time: 15–30 minutes per device type

Prerequisites

Before configuring video streaming, ensure:

- Devices are successfully added to GCXONE
- Network connectivity is established
- Device credentials are configured correctly
- Required ports are open: RTSP 554, HTTP 80, HTTPS 443
- For Local Mode: Operator workstations are prepared with network access

Configuration Workflow

1. Device-Level Streaming Configuration
2. GCXONE Platform Streaming Settings
3. Local Mode Setup (Optional)
4. Stream Quality Optimization
5. Verification

Part 1: Device-Level Streaming Configuration

Step 1: Configure Device Streaming Settings

Navigation: Device Management Interface → Video/Streaming Settings

1. Access the device's web interface or management software
2. Navigate to Video Settings or Streaming Configuration
3. Configure Primary Stream:
    - Resolution: Set appropriate resolution such as 1920x1080 or 1280x720
    - Frame Rate: 15–30 FPS recommended for most use cases
    - Bitrate: 2–4 Mbps typical based on network capacity
    - Codec: H.264 or H.265 — H.265 recommended for better compression
4. Configure Secondary Stream if available:
    - Resolution: Lower resolution such as 640x480

    - Bitrate: Lower bitrate for bandwidth optimization
    - Purpose: Used for multi-camera grid views

5. Enable RTSP Streaming if not already enabled
6. Set RTSP Port: Typically 554 or custom if required
7. Click Save or Apply

Note: Specific settings vary by device manufacturer. Refer to device documentation for exact navigation paths.

Step 2: Verify Network Connectivity

- From GCXONE platform, verify device shows Online status
- Test RTSP stream connection using the device IP and port 554
- Verify ports are accessible: RTSP on 554, HTTP on 80, HTTPS on 443
- Check firewall rules allow required ports

Part 2: GCXONE Platform Streaming Configuration

Step 3: Configure Camera Streaming Settings in GCXONE

Navigation: GCXONE → Devices → [Device Name] → Camera Configuration

1. Log into GCXONE web portal
2. Navigate to Devices and select your device
3. Click Configure Cameras or Camera Management
4. For each camera, configure streaming:

Cloud Streaming Settings:

- Enable Cloud Streaming for remote access
- Stream Source: Primary or Secondary stream
- Quality Mode: Auto, High, Medium, or Low
- Enable Adaptive Quality for bandwidth optimization

Local Streaming Settings:

- Enable Local Streaming if using Local Mode
- Stream Source: Primary stream recommended for local
- Quality Mode: High — local network can handle higher quality

1. Configure Advanced Streaming Options:
    - Buffer Size: Adjust for network conditions
    - Timeout Settings: Increase if experiencing frequent timeouts
    - Retry Logic: Configure automatic reconnection attempts
2. Click Save Configuration

Step 4: Configure Global Streaming Preferences

Navigation: GCXONE → Settings → Streaming Preferences

1. Configure Default Settings:
    - Default Quality: Auto recommended
    - Default Stream Source: Primary or Secondary
    - Concurrent Stream Limit: Maximum streams per user or workstation
2. Configure Bandwidth Management:
    - Enable Adaptive Quality
    - Set Bandwidth Threshold if required
    - Define Quality Degradation Policy
3. Configure Performance Settings:
    - Connection Timeout: 30–60 seconds recommended
    - Retry Attempts: 3–5 attempts recommended
4. Click Save Preferences

Part 3: Local Mode Setup (Optional but Recommended)

Step 5: Install Local Mode Service

Navigation: Operator Workstation → Genesis Installer

1. Download Genesis Installer for Local Mode
2. Run installer on operator workstation — requires admin privileges
3. Follow installation wizard:
    - Accept license agreement
    - Choose installation directory
    - Select Local Mode components and complete installation
4. Verify service is running in Windows Services
5. Restart operator workstation if required

Step 6: Configure Local Mode in GCXONE

Navigation: GCXONE → Settings → Local Mode Configuration

1. Enable Local Mode for supported devices
2. Enable Auto-Discovery for local network devices
3. Configure Local Mode Policies:
    - Enable Prefer Local Mode for operator workstations
    - Enable Fallback to Cloud for automatic fallback
    - Configure Local Network Detection with network ranges
4. Test Local Mode connectivity from operator workstation
5. Click Save Configuration

Part 4: Stream Quality Optimization

High-Priority Monitoring

- Quality: High
- Stream Source: Primary
- Use Local Mode if available

Multi-Camera Grid Views

- Quality: Medium to Low
- Stream Source: Secondary if available
- Enable Adaptive Quality and limit concurrent streams

Mobile or Remote Access

- Quality: Auto (adaptive)
- Stream Source: Secondary for lower bandwidth
- Enable bandwidth optimization and use cloud streaming

Recording or Playback

- Quality: High for primary recordings
- Stream Source: Primary
- Use device's recording settings

Part 5: Verification Checklist

Cloud Streaming

- Live video displays correctly in GCXONE
- Stream quality is acceptable
- Multiple cameras can stream simultaneously
- Streaming works from remote locations
- Adaptive quality adjusts based on bandwidth

Local Mode

- Local Mode service is running
- Direct connections work from operator workstation
- Lower latency compared to cloud streaming
- Automatic fallback to cloud works

Performance

- No excessive buffering or lag
- Stream starts within 2–3 seconds
- Multiple concurrent streams work smoothly
- Bandwidth usage is within expected ranges
- No connection drops during normal operation

Advanced Configuration

Multi-Stream Configuration

- Configure Primary Stream for high quality full-screen viewing
- Configure Secondary Stream for lower quality grid views
- Configure Sub-Stream for very low quality thumbnails
- Use stream selection based on viewing context

Bandwidth Throttling

- Enable Adaptive Quality for all cameras
- Set Bandwidth Limits in GCXONE settings
- Use Secondary Streams for grid views
- Limit Concurrent Streams per user

Failover Configuration

- Configure Primary and Backup Streams
- Enable Automatic Failover in device settings
- Set up Stream Redundancy in GCXONE
- Test failover scenarios regularly

Quick Troubleshooting

- No video: Check device status, verify credentials, test RTSP connection
- Poor quality: Adjust bitrate settings, check bandwidth, verify stream source
- Buffering: Reduce quality, check network latency, increase buffer size
- Connection drops: Check network stability, verify timeout settings, check firewall
- Local Mode not working: Verify service is running, check network access, verify device support